

2nd Progress Report

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Outcome of the previous meeting

- Discussed the topic of research questions, talked over the preliminary set of questions i have written.
- Shouldn't focus on/not much potential in applying RL for guidance and navigation, focus on control.
- Adviced to look at the demand side of the research field instead of the supply/what-other-MSc-have-done side.

Description of the results since last meeting

- Revisited research questions during the week a couple times, each with a fresh mind:

Research Objective:

To advance the state-of-the-art RL based fault tolerant flight controllers and further the technological readiness level of the Flying-V, by developing a reinforcement learning based intelligent flight control system for the Flying-V.

Research Questions:

1. What reinforcement learning algorithm has most promise to the problem of fault tolerant flight control?
 - (a) What RL algorithms are used in state-of-the-art flight control?
 - (b) What desirable qualities should an algorithm possess for the best fault tolerant controller?
 - (c) What preliminary comparison should be done to select the most promising algorithm?
2. What are the handling qualities or possible faults scenarios of the Flying-V that a controller should take into account?
3. How can the identified algorithm be applied to control the Flying-V?
 - (a) How are RL based flight controllers implemented?
 - (b) How will the algorithm interface with existing simulations of the Flying-V?
4. How does the implemented flight controller perform both in the presence of faults and in nominal flight?
 - (a) What criteria should be kept track of to characterize a controllers fault tolerance?

- (b) Noting the possible fault scenarios and handling qualities of the Flying-V, what flight scenarios should be designed to test the performance of the controller?
 - (c) How well does the nominal flight performance of the proposed flight controller compare to other intelligent and traditional controllers?
 - (d) How are the fault tolerance characteristics of the proposed flight controller compared to other intelligent and traditional controllers?
- Reviewed a batch of literature on the flying-V as well as flying wings and blended-wing-bodies, using sources from Scopus, from other MSc theses, from publications of the Roelof Vos and Benad.

Found conflicting information regarding handling quality of flying-v. E.g Thibaut claims dutch roll during cruise and approach is only unstable mode, while simon claims the same in addition to claiming phugoid and spiral to be unstable during approach. Commonality of both claims is the unstable dutch roll, which is enough of a reason to design an AFCS.

Literature on other flying wing designs also finds that lateral modes are where such designs are poor in, notably an unstable dutch roll mode. Since Willem did not try to create an RL controller for lateral control, and the Flying-V has unstable dutch roll, there is thus **a gap** in studies on performance of RL controllers in handling attitude control for Flying-V (specifically seeing how well they can stabilize dutch roll).

- Found the SHERPA algorithm which seems interesting since it can be used to avoid agents from executing unsafe actions during episodes. Read about how Tijmen Pollack in his MSc thesis used SHERPA to preserve *ergodicity*, which i understood as making sure that his RL agent did not use unsafe actions and crash during exploration.
- Found the IGHPa algorithm which seems interesting for 2 reasons.
 1. Previous online-offline implementations from Casper and Lucas used (D)SAC IDHP, which limited the utilization of offline-trained networks since (D)SAC's critic and IDHP's critic do not approximate the same functions¹, thus the critic network from (D)SAC is not reused when transferring networks between offline and online algorithms.
 2. IGDPHa is suggested to be slightly more powerful than IDHP from zhou's journal paper, in addition to GDHP being touted as more powerful than DHP or HDP.
- Gathered a couple of recent papers on reinforcement learning for flight control which i am in the process of reading.

Plans for next 2 weeks

1. Continue reading papers on reinforcement learning from the supply/more recent side, and iterating on research questions.
2. Perhaps look into Explainable AI work (reading other MSc theses first, and then broader literature search) and see if it would be feasible to incorporate some ideas.
3. Try to implement/play with code for an RL algorithm (ideally in a continuous state-action space) for more familiarity.

¹from what i gather, (D)SAC estimates value function but (I)DHP estimates gradient of value function